

2. Which is the chief heavenly body of solar system?

- (a) Earth (b) Jupiter
(c) Saturn (d) Sun
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (d)

The main celestial body of the solar system is sun. Sun is located at the centre of solar system.

iv. The Mercury

1. Which of the following planets does not have a satellite?

- (a) Mars (b) Mercury
(c) Neptune (d) Pluto

44th B.P.S.C. (Pre) 2000

42nd B.P.S.C. (Pre) 1997

Ans. (b)

There are only two planets in our solar system having no natural satellite which are - (i) Mercury (ii) Venus. The planet having satellite are as follows –

Earth-1 Mars-2
Jupiter-95 Saturn-146
Uranus-28 Neptune-16

In addition to these, Pluto (Dwarf Planet) has five (5) known satellites.

v. The Venus

1. The hottest planet of the solar system is –

- (a) Mercury (b) Venus
(c) Mars (d) Earth

41st B.P.S.C. (Pre) 1996

Ans. (b)

Average temperature of all Planet in our Solar System is as follows –

Mercury = 167° C
Venus = 464° C
Earth = 15° C
Mars = –65° C
Jupiter = –110° C
Saturn = –140° C
Uranus = –195° C
Neptune = –200° C

Thus, Venus is the hottest planet in our Solar System and Neptune is the coldest planet.

2. Which planet is considered as the Earth's twin planet due to similarity in shape and size?

- (a) Mars (b) Jupiter
(c) Venus (d) More than one of the above
(e) None of the above

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Ans. (c)

Planet Venus is considered as the Earth's twin planet due to the similarity in shape and size and having very similar composition.

3. Which planet is known as the 'Evening Star'?

- (a) Mars (b) Jupiter
(c) Venus (d) Saturn

44th B.P.S.C. (Pre) 2000

Ans. (c)

Venus is known as the Evening Star and also as the Morning Star while Mars is called the Red Planet and Saturn has lowest density among all planets (lower than water also).

vi. The Earth

1. Time taken by light emitted from sun to reach earth is –

- (a) 2 minutes (b) 1 minutes
(c) 8 minutes (d) 16 minutes

38th B.P.S.C. (Pre) 1992

Ans. (c)

The radiation from the sun which is popularly known as sunlight is a mixture of electromagnetic waves. Sunlight takes an average of 8 minutes and 19 seconds to travel from the Sun to the Earth. These electromagnetic energy waves travel with a speed of 3.0×10^{18} m/sec. Thus, the closest answer given in option (c) is true.

2. In completing one revolution of the sun, Earth takes approximately –

- (a) 365 days (b) 365.25 days
(c) 365.5 days (d) 365.75 days

41st B.P.S.C. (Pre) 1996

Ans. (b)

The Earth takes 365.25636 days or 365 days, 5 hours 59 minutes and 9.51 second to complete orbit around the Sun. This period is known as 'Solar Year'. The solar year is also known as The tropical year. In the year 2000, the increase of 5 Minute had been noted in comparison to other years. In this period the time was calculated to 365 days, 6 hours 13 minutes (in place of 9 minutes), 53.26 second (in place of 9.51 second). Such year is known as 'Anomalistic Year'.

3. Approximately how much distance per minute does earth cover while revolving?

- (a) 49 km (b) 59 km
(c) 69 km (d) 79 km

44th B.P.S.C. (Pre) 2001

Ans. (*)

Earth's equatorial perimeter is about 40,090 Km. This distance is covered by Earth in 24 hours and this indicates that cruising speed of Earth on its axis per hour is 40,090.28 Km/ 24 hour = 1670 Km/ hour (approx). According to this calculation the cruising speed per minute is = 1670 Km/ 60 min = 27.8 Km/ min. Thus, none of the given answers is correct.

Note : In question rotating should be on the place of revolving.

4. Who among the following was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the Sun?

- (a) Aryabhata (b) Bhaskara
(c) Brahmagupta (d) Varahamihira
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Aryabhata was the first to explain that the rotation of the earth on its own axis accounts for the daily rising and setting of the sun. His works are compiled in Aryasiddhanta. Bhaskara-I was a proponent of Bhedabheda School of Vedanta Philosophy. Varahamihira's notable works include Brihat Sanhita, Pancha Sididhantika.

5. The possibility of a desert on earth is more

- (a) Nearby 0° latitude (b) Nearby 23° latitude
(c) Nearby 50° latitude (d) Nearby 70° latitude

41st B.P.S.C. (Pre) 1996

Ans. (b)

Desert climate is also Known as arid climate. The perfect climate for the desert is Tropical climate which is spread mostly between 15°-30° between North and South of the Equator. There are four main geographical situations for the formation of desert –

- (1) The presence of high air pressure near 30° latitude. Sahara and Australia desert fall under this climate.
(2) The western coastal side of continent found between 20°-30° latitude. Here cold oceanic current is reason for desert formation. Arizona desert, Mexican desert, Atacama desert, Namibian desert come under this climate.

- (3) Death Valley (North America), Patagonian desert (Argentina) and Peruvian desert are the deserts found in mountains rain shadow region.
(4) The desert situated in the mid part of the Continents, far from the humid air. Gobi desert (China and Mongolia), Australian desert and Great basin desert (USA) come under this group.

6. The animals found in desert are called

- (a) Deserticolous
(b) Desertimalia
(c) Desertimals
(d) More than one of the above
(e) None of the above

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Ans. (a)

The term 'deserticolous' refers to organisms that inhabit desert regions. Animals in deserts are exposed to limited quantities of food, low and unpredictable water supplies, high and widely varying temperatures, loose substrates, high incident radiation, and lack of concealment because of scant vegetation.

7. Geodesy is the science which deals with –

- (a) The dating of terrestrial rock.
(b) Measurement of Earth's dimensions
(c) Measurement of elevation and depression of the earth
(d) Recording of changes made by the crust
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

Geodesy is the science of accurately measuring and understanding the three fundamental properties of the Earth. These properties are the geometric shape of the earth, the orientation of the earth in space and the gravitational field of the earth; along with it also studies the changes in the above properties with time. Thus geodesy is the science which deals with the measurement of the dimensions of the earth.

8. When a bar magnet is placed in the earth's magnetic field, with its north pole pointing towards the geographical north, the number of neutral points around the magnet is

- (a) 2 (b) 4
(c) 6 (d) 5
(e) None of the above / More than one of the above

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Ans. (a)